ANDHIKA WICAKSONO SASONGKO

Water Resources Engineer/Hydrologist

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PROFESSIONAL SUMMARY

Water Resources Engineer with 5 year experience in hydrology, hydraulic modeling, dam safety assessment, hydropower development, river engineering, mining water management, and raw water supply infrastructure. Proven expertise in design flood estimation, water availability analysis, flood forecasting systems, river diversion projects, and hydrological studies for dams, weirs, hydropower facilities, and mining infrastructure. Experienced in Dam Safety Assessment projects involving as well as ADB-funded water resources programs. Strong capability in engineering studies, technical design reviews, and multidisciplinary project coordination for sustainable water and energy infrastructure developments.

EDUCATION

Master of Water Resources Engineering and Management

Bandung Institute of Technology (ITB)
2021 – 2023

Thesis:
Estimation of Economic Damage Valuation in Gorontalo City Caused by Flood from Bone River

Bachelor of Water Resources Engineering and Management

Bandung Institute of Technology (ITB)
2017 – 2021

Final Project:
Planning of Lowland Irrigation Water Management at Block A5 Food Estate Area Dadahup

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CORE COMPETENCIES

- Hydrology & Water Resources Assessment
- Hydraulic Modeling & River Engineering
- Dam Safety Assessment, Hydropower & Water Infrastructure
- Mining Water Management & Surface Water Systems
- Flood Forecasting, Flood Risk & Climate Resilience

SELECTED PROJECT EXPERIENCE

Flood Forecasting Early Warning System (FFEWS) – Ambon City (2024 - 2025)

Role: Hydrologist

- Developed flood forecasting models and early warning systems.
- Conducted rainfall-runoff analysis and hydrological modeling using Python.
- Supported implementation of an ADB-funded climate resilience and disaster risk reduction project.

Floating Solar Photovoltaic (FPV) Assessment – Duriangkang and Tembesi Reservoirs (2025)

Role: Hydrologist

- Conducted hydrological assessments to support feasibility studies of floating solar photovoltaic (FPV) installations.
- Evaluated historical and projected reservoir water level fluctuations.
- Assessed reservoir storage behavior and water balance conditions relevant to FPV deployment.
- Analyzed potential hydrological impacts of photovoltaic coverage on reservoir operation and water resource availability.
- Supported integration of renewable energy infrastructure with existing reservoir functions and water supply objectives.

Development of Automatic Weather Station (AWS) for PT Aimtopindo Nuansa Kimia

Role: Hydrologist

- Developed the hydrometeorological monitoring concept to support operational water management.
- Determined the required meteorological and hydrological parameters, including rainfall, air temperature, relative humidity, wind speed and direction, solar radiation, and atmospheric pressure.
- Recommended sensor specifications and installation locations based on site conditions and monitoring objectives.
- Supported data acquisition and telemetry planning for continuous environmental monitoring.
- Conducted hydrological analysis using observed weather data to support water resources assessment and operational decision-making.

Development of Raceway Pond for CO_2 Capture Algae System

Role: Hydraulic Engineer

- Designed the hydraulic configuration of a raceway pond for algae cultivation and CO_2 capture applications.

- Performed Computational Fluid Dynamics (CFD) simulations using ANSYS Fluent to evaluate flow characteristics within the raceway.
- Optimized paddlewheel operating conditions to achieve the target flow velocity and ensure uniform circulation throughout the pond.
- Evaluated hydraulic performance, mixing efficiency, and flow distribution to improve algae growth conditions.
- Provided engineering recommendations for paddlewheel configuration and hydraulic system optimization.

Wae Lega Small Hydropower Project (2023)

Role: Hydrologist

- Conducted hydrological assessment for small hydropower development.
- Performed dependable flow and water availability analyses.
- Reviewed hydrological design parameters supporting project development.

Dam Safety Assessment Projects (2022 – Present)

Role: Hydrologist

Pamukkulu Dam

- Conducted design flood estimation and hydrological safety evaluation.
- Performed extreme rainfall and flood analyses to support dam safety review.
- Performed water availability analysis for Dam Operation Manual

Rempang Dam

- Conducted hydrological analyses for dam safety assessment requirements.
- Evaluated design flood parameters and reservoir inflow characteristics.
- Performed low flow analysis to support the determination of dam firm yield.

Mukakuning Dam, Duriangkang Dam, and Sei Harapan Dam

- Performed hydrological reviews of dam infrastructure systems.
- Evaluated catchment hydrology, design rainfall, and flood characteristics.
- Supported technical assessments related to dam operation and safety.

Kedungombo Dam

- Conducted hydrological assessment and design flood evaluation.
- Supported review of hydrological parameters for dam safety compliance.
- Conducted reservoir sedimentation study and estimated reservoir useful life based on sediment inflow and storage loss analysis.

Cacaban Dam

- Performed hydrological analysis and flood estimation studies.

- Evaluated hydrological inputs required for dam safety assessment.

Riko Weir Raw Water Supply Project – Nusantara Capital City (2025)

Role: Water Resources Engineer

- Supported Detail Engineering Design (DED) of Riko Weir for raw water supply infrastructure.
- Conducted water availability assessment and dependable flow analysis.
- Evaluated hydrological characteristics of the catchment area.
- Supported planning for long-term raw water demand fulfillment.

Binungan River Diversion Project (2022)

Role: Hydrology Engineer

- Conducted hydrological and hydraulic analysis for river diversion design.
- Performed design flood estimation and channel capacity assessment.
- Evaluated diversion channel performance under various flow conditions.

Ninian River Diversion Project (2021)

Role: Hydrology Engineer

- Conducted flood hydrology analysis and hydraulic assessments.
- Evaluated river diversion alternatives and hydraulic performance.
- Supported surface water management planning.

PMSS Surface Water Management Project (2021)

Role: Hydrology Engineer

- Developed mine water management strategies under Life of Mine (LOM) conditions.
- Conducted runoff analysis and water balance assessment.
- Supported integrated hydro-geotechnical studies.

RESEARCH & PUBLICATIONS

Economic Loss due to the Failure of a Cascade Dam: A Case Study of the Saguling–Cirata–Jatiluhur Dam System

2024

Evaluation of Sandtrap Structure Performance on Tami Weir and the Effect of Sediment Accumulation on Irrigation Service

Accepted for Publication

ACADEMIC EXPERIENCE

Assistant Lecturer

Bandung Institute of Technology (ITB)
2018 – 2021

Courses:

- Hydrology I
- Hydrology II
- Hydraulics
- Water Resources Infrastructure

CERTIFICATIONS

Associate Class Expert of Water Resources Engineering (Ahli Madya)

BNSP | 2023

ACHIEVEMENTS

1st Runner-Up – Dam Innovation Contest

Diponegoro University | 2020